Information Security

Lecture 3

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Announcements:

Four quizzes: Best three. No make up quizzes. 30%

Two assignments: Both 10% (total 20%)

One midterm: 25%

One final exam: 25%

Verification of Security Protocols

Abstract the protocol into a **formal model** (automata, logic)

Assume perfect cryptography

$$A \rightarrow B : \{(A, \{m\}_B)\}_B$$

$$B \rightarrow A : \{(B, \{m\}_A)\}_A$$

Specify required guarantees as **mathematical properties** over these abstract models

I has access to m.

Prove these properties hold, preferably by automated means

?

Our Formalism: Dolev-Yao Model

On the Security of Public Key Protocols (1983).

$$A \rightarrow B : \left\{ \left(A, \{m\}_B \right) \right\}_B$$

$$B \rightarrow A : \left\{ \left(B, \{m\}_A \right) \right\}_A$$

$$A!B : \mathrm{aenc}\left(\left(A, \mathrm{aenc}\left(m, \mathrm{pk}_B \right) \right), \mathrm{pk}_B \right)$$

$$B? : \mathrm{aenc}\left(\left(X, \mathrm{aenc}\left(m', \mathrm{pk}_B \right) \right), \mathrm{pk}_B \right)$$

$$B!X : \mathrm{aenc}\left(\left(B, \mathrm{aenc}\left(m', \mathrm{pk}_X \right) \right), \mathrm{pk}_X \right)$$

$$A? : \mathrm{aenc}\left(\left(B, \mathrm{aenc}\left(m, \mathrm{pk}_A \right) \right), \mathrm{pk}_A \right)$$

Our Formalism: Dolev-Yao Model

Split each communication into a send and a receive.

The intruder I is essentially the network.

- Each send captured by I
- Each receive assumed to come from I

A send action need not have a corresponding receive action.

Our Formalism: Dolev-Yao Model

Intruder I cannot break encryption. It can:

- **See** any message sent on the public channel
- **Block** any message from reaching the intended recipient
- **Re-route** any message to any principal
- **Masquerade** as any principal and send messages in their name
- **Initiate** new communication according to the protocol
- **Generate** messages according to some rules

Messages as Term Algebra

Messages are **not** structured documents.

Ignore extraneous details (headers, metadata, formatting)

Formally modelled as symbolic terms

$$t := m \mid a \mid \mathrm{pk}(a) \mid \mathrm{sk}(a) \mid \mathrm{pair}(t_1, t_2) \mid \mathrm{aenc}(t, \mathrm{pk}(a))$$

Atomic terms m (messages) and a (agent names).

Proof Rules for Generating Terms

$$\frac{}{X \vdash t}\ \text{ax}\quad (t \in X)$$

$$\frac{X \vdash a}{X \vdash \text{pk}(a)}\ \text{pk}$$

$$\frac{X \vdash \text{pair}(t_0, t_1)}{X \vdash t_i}\ \text{split}_i$$

$$\frac{X \vdash t \qquad X \vdash t'}{X \vdash \text{pair}(t, t')}\ \text{pair}$$

$$\frac{X \vdash \text{aenc}(m, \text{pk}(a)) \qquad X \vdash \text{sk}(a)}{X \vdash m}\ \text{adec}$$

$$\frac{X \vdash m \qquad X \vdash \text{pk}(a)}{X \vdash \text{aenc}(m, \text{pk}(a))}\ \text{aenc}$$

Proof Rules for Generating Terms

$$
\dfrac{\dfrac{\dfrac{\dfrac{\dfrac{}{X_I \vdash I}\ \text{ax}\ (I \in X_I) \qquad \dfrac{\dfrac{X_I \vdash \text{aenc}\left(\text{pair}\left(B, \text{aenc}\left(\text{pair}\left(A, \text{aenc}\left(m, \text{pk}(B)\right)\right), \text{pk}(I)\right)\right), \text{pk}(I)\right)}{X_I \vdash \text{pair}\left(B, \text{aenc}\left(\text{pair}\left(A, \text{aenc}\left(m, \text{pk}(B)\right)\right), \text{pk}(I)\right)\right)}\ \text{split}_1 \qquad \dfrac{}{X_I \vdash \text{sk}(I)}\ \text{ax}\ (\text{sk}_I \in X_I)}{X_I \vdash \text{aenc}\left(\text{pair}\left(A, \text{aenc}\left(m, \text{pk}(B)\right)\right), \text{pk}(I)\right)}\ \text{adec}}{X_I \vdash \text{pair}\left(A, \text{aenc}\left(m, \text{pk}(B)\right)\right)}\ \text{split}_1}{X_I \vdash \text{aenc}\left(m, \text{pk}(B)\right)}\ \text{pair} \qquad \dfrac{\dfrac{}{X_I \vdash B}\ \text{ax}\ (B \in X_I)}{X_I \vdash \text{pk}(B)}\ \text{pk}}{X_I \vdash \text{pair}\left(I, \text{aenc}\left(m, \text{pk}(B)\right)\right)}\ \text{aenc}}{X_I \vdash \text{aenc}\left(\text{pair}\left(I, \text{aenc}\left(m, \text{pk}(B)\right)\right), \text{pk}(B)\right)}
$$

Proof Rules for Generating Terms

Given I's knowledge, can it derive a given term t?

$$\Longleftrightarrow$$

Given a deductive proof system for generating terms and a set of known terms X, *does there exist a derivation for term* t?

That is, $X \vdash t$?

And can we automate this?

Two Intruder Problems

Passive intruder problem: Can the intruder violate some property just by listening to the network?

Active intruder problem: Can the intruder violate some property by listening to network + allowable behaviours?

Two Intruder Problems

Passive intruder problem: Can the intruder violate some property just by listening to the network?

Fixed X, fixed t. Check if $X \vdash t$.

Active intruder problem: Can the intruder violate some property by listening to network + allowable behaviours?

Two Intruder Problems

Passive intruder problem: Can the intruder violate some property just by listening to the network?

Fixed X, fixed t. Check if $X \vdash t$.

Active intruder problem: Can the intruder violate some property by listening to network + allowable behaviours?

Come up with an X and a suitable mapping for variables in X and t such that we have $X \vdash t$.

Passive Intruder Problem

Given X and t, check if $X \vdash t$.

Passive Intruder Problem

Given X and t, check if $X \vdash t$.

Consider

$$X = \{\mathrm{aenc}(m, \mathrm{pk}(A)),$$
$$\mathrm{pair}(\mathrm{sk}(B), \mathrm{aenc}(\mathrm{pair}(n, \mathrm{sk}(A)), \mathrm{pk}(C)))$$
$$\mathrm{pair}(n, \mathrm{aenc}(\mathrm{sk}(C), \mathrm{pk}(B)))\}$$

$$X \vdash m?$$

Passive Intruder Problem

Given X and t, check if $X \vdash t$.

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$$X = \{\mathrm{aenc}(m, \mathrm{pk}(A)),$$
$$\mathrm{pair}(\mathrm{sk}(A), \mathrm{aenc}(\mathrm{pair}(n, \mathrm{sk}(A)), \mathrm{pk}(C)))$$
$$\mathrm{pair}(n, \mathrm{aenc}(\mathrm{sk}(C), \mathrm{pk}(B)))\}$$

$$X \vdash m?$$

Passive Intruder Problem

Given X and t, check if $X \vdash t$.

Consider

$$X = \{\mathrm{aenc}(m, \mathrm{pk}(A)),$$
$$\mathrm{pair}(\mathrm{pk}(B), \mathrm{aenc}(\mathrm{pair}(m, \mathrm{pk}(B)), \mathrm{pk}(C)))$$
$$\mathrm{aenc}(\mathrm{sk}(C), \mathrm{pk}(B))\}$$

$$X \vdash \mathrm{aenc}(m, \mathrm{pk}(C))?$$

Automated Proof Discovery

Given X and t, check if $X \vdash t$.

Rules do not change X.

Some rules *construct*, that is give rise to bigger terms: encryption, pairing

Some rules *destruct*, that is give rise to smaller terms: decryption, split

Automated Proof Discovery

What is the size of a term?

Rules do not change X.

Some rules *construct*, that is give rise to **bigger terms**: encryption, pairing

Some rules *destruct*, that is give rise to **smaller terms**: decryption, split

Size of Terms

$$\mathrm{aenc}\left(\mathrm{pair}\left(B, \mathrm{aenc}\left(\mathrm{pair}\left(A, \mathrm{aenc}\left(m, \mathrm{pk}(B)\right)\right), \mathrm{pk}(I)\right)\right), \mathrm{pk}(I)\right)$$

Treat a term like a tree. Count the number of nodes!

Formally modelled as symbolic terms

$$t := m \;\mid\; a \;\mid\; \mathrm{pk}(a) \;\mid\; \mathrm{sk}(a) \;\mid\; \mathrm{pair}(t_1, t_2) \;\mid\; \mathrm{aenc}(t, \mathrm{pk}(a))$$

Size of Terms

$$\mathrm{aenc}\left(\mathrm{pair}\left(B, \mathrm{aenc}\left(\mathrm{pair}\left(A, \mathrm{aenc}\left(m, \mathrm{pk}(B)\right)\right), \mathrm{pk}(I)\right)\right), \mathrm{pk}(I)\right)$$

$$\mathrm{size}(t) = \begin{cases} 1 & \text{if } t \text{ is atomic} \\ 1 + \sum \mathrm{size}(t_i) & \text{if } t = f(t_1, t_2, \cdots, t_k) \end{cases}$$

Formally modelled as symbolic terms

$$t := m \;\mid\; a \;\mid\; \mathrm{pk}(a) \;\mid\; \mathrm{sk}(a) \;\mid\; \mathrm{pair}(t_1, t_2) \;\mid\; \mathrm{aenc}(t, \mathrm{pk}(a))$$

Abnormal Proofs

$$\dfrac{\dfrac{X \vdash t_0 \qquad X \vdash t_1}{X \vdash \mathrm{pair}(t_0, t_1)}\,\mathrm{pair}}{X \vdash t_0}\,\mathrm{split}_0$$

$$\dfrac{\dfrac{X \vdash m \qquad X \vdash \mathrm{pk}(a)}{X \vdash \mathrm{aenc}(m, \mathrm{pk}(a))}\,\mathrm{aenc} \qquad X \vdash \mathrm{sk}(a)}{X \vdash m}\,\mathrm{adec}$$

One where a *construct rule* is immediately followed by *destruct rule*.

Normal Proofs

One where no *construct rule* is immediately followed by *destruct rule*.

Normal Proofs

A normal proof is one where the major premise of a *destructor rule* is not obtained by the application of a *constructor rule*.

One where no *construct rule* is immediately followed by *destruct rule*.

Normal Proofs

destructor rule

$$\frac{}{X \vdash t}\ \text{ax}\quad (t \in X)$$

$$\frac{X \vdash \text{pair}(t_0, t_1)}{X \vdash t_i}\ \text{split}_i$$

$$\frac{X \vdash \text{aenc}(m, \text{pk}(a)) \quad X \vdash \text{sk}(a)}{X \vdash m}\ \text{adec}$$

constructor rule

$$\frac{X \vdash a}{X \vdash \text{pk}(a)}\ \text{pk}$$

$$\frac{X \vdash t \quad X \vdash t'}{X \vdash \text{pair}(t, t')}\ \text{pair}$$

$$\frac{X \vdash m \quad X \vdash \text{pk}(a)}{X \vdash \text{aenc}(m, \text{pk}(a))}\ \text{aenc}$$

Normal Proofs

If there exists a proof of $X \vdash t$, then there exists a minimal proof.

The shortest proof π must be normal.

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If there exists a proof of $X \vdash t$, then there exists a minimal proof.

The shortest proof π must be normal.

Suppose not!

Then there is a subproof ξ of $X \vdash u$ such that ξ ends in a destructor rule, and the major premise of ξ is yielded by some constructor rule. We will show how to replace ξ by a smaller proof of $X \vdash u$, thus contradicting the minimality of π.

Normal Proofs

If there exists a proof of $X \vdash t$, then there exists a minimal proof.

The shortest proof π must be normal.

Case I:

$$\frac{}{X \vdash t}\ \mathrm{ax}\ \ (t \in X)$$

major premise is empty.

Normal Proofs

If there exists a proof of $X \vdash t$, then there exists a minimal proof.

The shortest proof π must be normal.

Case II:

$$\dfrac{\dfrac{\begin{array}{c}\pi_0\\ \vdots\\ X \vdash t_0\end{array} \qquad \begin{array}{c}\pi_1\\ \vdots\\ X \vdash t_1\end{array}}{X \vdash \mathrm{pair}(t_0, t_1)}\ \mathrm{pair}}{X \vdash t_i}\ \mathrm{split}_i$$

Normal Proofs

If there exists a proof of $X \vdash t$, then there exists a minimal proof.

The shortest proof π must be normal.

Case II:

$$\dfrac{\dfrac{\begin{array}{c}\pi_0\\ \vdots\\ X \vdash t_0\end{array} \qquad \begin{array}{c}\pi_1\\ \vdots\\ X \vdash t_1\end{array}}{X \vdash \mathrm{pair}(t_0, t_1)}\ \mathrm{pair}}{X \vdash t_i}\ \mathrm{split}_i \quad \Rightarrow \quad \begin{array}{c}\pi_i\\ \vdots\\ X \vdash t_i\end{array}$$

Normal Proofs

If there exists a proof of $X \vdash t$, then there exists a minimal proof.

The shortest proof π must be normal.

Case III:

$$\dfrac{\dfrac{\begin{array}{c}\pi' \\ \vdots \\ X \vdash m\end{array} \qquad \begin{array}{c}\pi'' \\ \vdots \\ X \vdash \mathrm{pk}(a)\end{array}}{X \vdash \mathrm{aenc}(m, \mathrm{pk}(a))}\,\mathrm{aenc} \qquad \begin{array}{c}\pi''' \\ \vdots \\ X \vdash \mathrm{sk}(a)\end{array}}{X \vdash m}\,\mathrm{adec} \quad \Rightarrow \quad \begin{array}{c}\pi' \\ \vdots \\ X \vdash m\end{array}$$

Normal Proofs

normalisation theorem

There exists a proof of $X \vdash t$ if and only if there exists a normal proof of $X \vdash t$.

Normal Proofs

normalisation theorem

There exists a proof of $X \vdash t$ if and only if there exists a normal proof of $X \vdash t$.

What is the size of the shortest normal proof?

Subterms

Treat a term like a tree. Subterm is like a subtree.

Formally modelled as symbolic terms

$$t := m \;\mid\; a \;\mid\; \mathrm{pk}(a) \;\mid\; \mathrm{sk}(a) \;\mid\; \mathrm{pair}(t_1, t_2) \;\mid\; \mathrm{aenc}(t, \mathrm{pk}(a))$$

Subterms

Treat a term like a tree. Subterm is like a subtree.

$$X \subseteq \mathrm{st}(X)$$

$$\mathrm{pair}(t_0, t_1) \in \mathrm{st}(X) \implies \{t_0, t_1\} \subset \mathrm{st}(X)$$

$$\mathrm{aenc}(m, k) \in \mathrm{st}(X) \implies \{m, k\} \subset \mathrm{st}(X)$$

Formally modelled as symbolic terms

$$t := m \mid a \mid \mathrm{pk}(a) \mid \mathrm{sk}(a) \mid \mathrm{pair}(t_1, t_2) \mid \mathrm{aenc}(t, \mathrm{pk}(a))$$

Subterms

Treat a term like a tree. Subterm is like a subtree.

$$\mathrm{aenc}\left(\mathrm{pair}\left(B, \mathrm{aenc}\left(\mathrm{pair}\left(A, \mathrm{aenc}\left(m, \mathrm{pk}(B)\right)\right), \mathrm{pk}(I)\right)\right), \mathrm{pk}(I)\right)$$

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$$\mathrm{pair}\left(B, \mathrm{aenc}\left(\mathrm{pair}\left(A, \mathrm{aenc}\left(m, \mathrm{pk}(B)\right)\right), \mathrm{pk}(I)\right)\right) \qquad \mathrm{pk}(I)$$

Subterms

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$$\mathrm{aenc}\left(\mathrm{pair}\left(A, \mathrm{aenc}\left(m, \mathrm{pk}(B)\right)\right), \mathrm{pk}(I)\right) \qquad B$$

Subterms

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$$\mathrm{pair}\left(B, \mathrm{aenc}\left(\mathrm{pair}\left(A, \mathrm{aenc}\left(m, \mathrm{pk}(B)\right)\right), \mathrm{pk}(I)\right)\right) \qquad \mathrm{pk}(I)$$

$$\mathrm{aenc}\left(\mathrm{pair}\left(A, \mathrm{aenc}\left(m, \mathrm{pk}(B)\right)\right), \mathrm{pk}(I)\right) \qquad B \qquad A$$

$$\mathrm{pair}\left(A, \mathrm{aenc}\left(m, \mathrm{pk}(B)\right)\right) \qquad \mathrm{aenc}\left(m, \mathrm{pk}(B)\right) \qquad m \qquad \mathrm{pk}(B)$$

Subterms

Treat a term like a tree. Subterm is like a subtree.

Is the number of subterms equal to the size of the term?

Subterms

Treat a term like a tree. Subterm is like a subtree.

Is the number of subterms equal to the size of the term?

$$|\,\mathrm{st}(X)\,| \leq \sum_{t \in X} |\,t\,|$$

Normal Proofs

There exists a proof of $X \vdash t$ if and only if there exists a normal proof of $X \vdash t$.

Claim:

Let π be a normal derivation of $X \vdash t$. If $X \vdash u$ is an intermediate step in $X \vdash t$, then $u \in \mathrm{st}(X \cup \{t\})$.

If the last rule is a destruction rule, then $u \in \mathrm{st}(X)$.

Subterm Property

Let π be a normal derivation of $X \vdash t$. If $X \vdash u$ is an intermediate step in $X \vdash t$, then $u \in \mathrm{st}(X \cup \{t\})$.

If the last rule is a destruction rule, then $u \in \mathrm{st}(X)$.

By Induction:

If the last term is generated by ax, then ...

Subterm Property

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By Induction:

If the last term is generated by pair, then ...

Subterm Property

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If the last rule is a destruction rule, then $u \in \mathrm{st}(X)$.

By Induction:

If the last term is generated by pair, then ...

$$\dfrac{\begin{array}{c}\pi_0 \\ \vdots \\ X \vdash t_0\end{array} \qquad \begin{array}{c}\pi_1 \\ \vdots \\ X \vdash t_1\end{array}}{X \vdash \mathrm{pair}(t_0, t_1)}\ \mathrm{pair}$$

Subterm Property

If $X \vdash u$ is an intermediate step...

Either it is in π_0 or π_1 or $u = \mathrm{pair}(t_0, t_1)$.

Therefore $u \in \mathrm{st}(X \cup \{t_0\}) \cup \mathrm{st}(X \cup \{t_0\}) \cup \{\mathrm{pair}(t_0, t_1)\}$

$$\dfrac{\begin{array}{c}\pi_0 \\ \vdots \\ X \vdash t_0\end{array} \qquad \begin{array}{c}\pi_1 \\ \vdots \\ X \vdash t_1\end{array}}{X \vdash \mathrm{pair}(t_0, t_1)}\ \mathrm{pair}$$

Subterm Property

If $X \vdash u$ is an intermediate step...

Either it is in π_0 or π_1 or $u = \mathrm{pair}(t_0, t_1)$.

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$$\dfrac{\begin{array}{c}\pi_0 \\ \vdots \\ X \vdash t_0\end{array} \qquad \begin{array}{c}\pi_1 \\ \vdots \\ X \vdash t_1\end{array}}{X \vdash \mathrm{pair}(t_0, t_1)}\ \mathrm{pair}$$

Subterm Property

Let π be a normal derivation of $X \vdash t$. If $X \vdash u$ is an intermediate step in $X \vdash t$, then $u \in \mathrm{st}(X \cup \{t\})$.

If the last rule is a destruction rule, then $u \in \mathrm{st}(X)$.

By Induction: If the last term is generated by encryption, then ...

$$\dfrac{\begin{array}{c}\pi' \\ \vdots \\ X \vdash m\end{array} \qquad \begin{array}{c}\pi'' \\ \vdots \\ X \vdash \mathrm{pk}(a)\end{array}}{X \vdash \mathrm{aenc}(m, \mathrm{pk}(a))}\ \mathrm{aenc}$$

Subterm Property

Let π be a normal derivation of $X \vdash t$. If $X \vdash u$ is an intermediate step in $X \vdash t$, then $u \in \mathrm{st}(X \cup \{t\})$.

If the last rule is a destruction rule, then $u \in \mathrm{st}(X)$.

By Induction:

If the last term is generated by split, then ...

$$\dfrac{\begin{array}{c}\pi \\ \vdots \\ X \vdash \mathrm{pair}(t_0, t_1)\end{array}}{X \vdash t_i}\ \mathrm{split}_i$$

Subterm Property

Let π be a normal derivation of $X \vdash t$. If $X \vdash u$ is an intermediate step in $X \vdash t$, then $u \in \mathrm{st}(X \cup \{t\})$.

If the last rule is a destruction rule, then $u \in \mathrm{st}(X)$.

By Induction: If the last term is generated by decryption, then ...

$$\dfrac{\begin{array}{c}\pi' \\ \vdots \\ X \vdash \mathrm{aenc}(m, \mathrm{pk}(a))\end{array} \qquad \begin{array}{c}\pi'' \\ \vdots \\ X \vdash \mathrm{sk}(a)\end{array}}{X \vdash m}\ \mathrm{adec}$$

Normalisation + Subterm Property

If there exists a proof of $X \vdash t$, if and only if there exists a normal proof of $X \vdash t$.

Let π be a normal derivation of $X \vdash t$. If $X \vdash u$ is an intermediate step in $X \vdash t$, then $u \in \mathrm{st}(X \cup \{t\})$.

If the last rule is a destruction rule, then $u \in \mathrm{st}(X)$.

Normalisation + Subterm Property

There exists a proof of $X \vdash t$ if and only if there exists a normal proof of $X \vdash t$, with each branch bounded by $|\mathrm{st}(X \cup \{t\})|$.

Normalisation + Subterm Property

There exists a proof of $X \vdash t$ if and only if there exists a normal proof of $X \vdash t$, with each branch bounded by the size of $X \cup \{t\}$, that is

$$|t| + \sum_{t' \in X} |t'|.$$

Naive Algorithm:

Let $N = |t| + \sum_{t' \in X} |t'|.$

If you have X, you can build

$X' = \{\text{terms generated from } X \text{ in one step}\}$

How?

Naive Algorithm:

Let $N = |t| + \sum_{t' \in X} |t'|$.

If you have X, you can build

$X' = \{$terms generated from X in one step$\}$

How?

$$X' = \{t : \exists \mathrm{pair}(t, _) \in X \vee \mathrm{pair}(_, t,) \in X\}$$

Naive Algorithm:

Let $N = |t| + \sum_{t' \in X} |t'|$.

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How?

$$X' = \{t : \exists \mathrm{pair}(t, _) \in X \vee \mathrm{pair}(_, t,) \in X\} \cup \{\mathrm{pair}(t, t') : t, t' \in X\}$$

Naive Algorithm:

Let $N = |t| + \sum_{t' \in X} |t'|$.

If you have X, you can build

$X' = \{\text{terms generated from } X \text{ in one step}\}$

How?

$$X' = \{t : \exists \mathrm{pair}(t, _) \in X \vee \mathrm{pair}(_, t,) \in X\} \cup \{\mathrm{pair}(t, t') : t, t' \in X\}$$
$$\cup \{m : \mathrm{sk}(A) \in X \wedge \mathrm{enc}(m, \mathrm{pk}(A)) \in X\}$$

Naive Algorithm:

Let $N = |t| + \sum_{t' \in X} |t'|$.

If you have X, you can build

$X' = \{$terms generated from X in one step$\}$

How?

$$X' = \{t : \exists \mathrm{pair}(t, _) \in X \vee \mathrm{pair}(_, t,) \in X\} \cup \{\mathrm{pair}(t, t') : t, t' \in X\}$$
$$\cup \{m : \mathrm{sk}(A) \in X \wedge \mathrm{enc}(m, \mathrm{pk}(A)) \in X\}$$
$$\cup \{\mathrm{enc}(m, \mathrm{pk}(A)) : m \in X, \mathrm{pk}(A) \in X\} \cup \{\mathrm{pk}(A) : A \in X\}$$

Naive Algorithm:

Let $N = |t| + \sum_{t' \in X} |t'|$.

If you have X, you can build

$X' = \{$terms generated from X in one step$\}$

In time $O(|X|^2)$.

Naive Algorithm:

Let $N = |t| + \sum_{t' \in X} |t'|$.

If you have X, you can build

$X' = \{$terms generated from X in one step$\}$

In time $O(N^2)$.

Naive Algorithm:

Let $N = |t| + \sum_{t' \in X} |t'|$.

If you have X, you can build

$X' = \{$terms generated from X in one step$\}$

In time $O(N^2)$.

Repeat this N times.

Naive Algorithm:

Let $N = |t| + \sum_{t' \in X} |t'|$.

In time $O(N^3)$, construct the set of all terms derivable in at most N steps.

t derivable in N steps $\iff X \vdash t$.

Naive Algorithm:

Let $N = |t| + \sum_{t' \in X} |t'|$.

Inputs: X, t

Can you do it
in linear time?

For i in range $(0,N)$,

Construct X' in $O(N^2)$

Set $X \leftarrow X'$

Check $t \in X$!

Passive Intruder Problem:

Passive intruder problem: Can the intruder violate some property just by listening to the network?

Fixed X, fixed t. Check if $X \vdash t$.

Decidable in polynomial time given certain operators and term algebra.

Passive Intruder Problem:

Passive intruder problem: Can the intruder violate some property just by listening to the network?

Fixed X, fixed t. Check if $X \vdash t$.

Decidable in polynomial time given certain operators and term algebra.

$$t := m \mid a \mid \mathrm{pk}(a) \mid \mathrm{sk}(a) \mid \mathrm{pair}(t_1, t_2) \mid \mathrm{aenc}(t, \mathrm{pk}(a))$$
$$k \mid \mathrm{senc}(t, k) \mid \mathrm{sign}(t, \mathrm{sk}(a)) \mid \mathrm{hash}(t)$$

Active Intruder Problem:

Active intruder problem: Can the intruder violate some property by listening to network + allowable behaviours?

$$A \rightarrow \quad : \left(A, \mathrm{aenc}(m, \mathrm{pk}_B)\right)$$

$$I \rightarrow B : \left(I, \mathrm{aenc}(m, \mathrm{pk}_B)\right)$$

$$B \rightarrow I : (B, \mathrm{aenc}(m, \mathrm{pk}_I))$$

$$\rightarrow A : (B, \mathrm{aenc}(m, \mathrm{pk}_A))$$

Active Intruder Problem:

- Take an arbitrary Turing machine M
- Encode its configurations (state, tape, head position) as symbolic messages
- Design a protocol such that:
  - Each valid protocol step corresponds to one transition of M.
  - The intruder can drive the protocol forward iff M makes a valid transition
- Define an insecure state iff M reaches a halting configuration.

Formal Verification Tools

DY-based bounded Active Intruder detection.

Relational modelling with a SAT solver.

ProVerif

Tamarin

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What all can you verify?

As models:

Dolev–Yao adversary

- Full network control (intercept, replay, modify, inject)
- Perfect cryptography (no guessing / no breaking primitives)

Multiple concurrent protocol sessions

Compromised principals (key reveal, corruption models)

What all can you verify?

As specifications:

Secrecy/Confidentiality: Message secrecy, key secrecy

Authentication: Aliveness, weak agreement

Protocol Correctness: Message origin authenticity, session binding, freshness guarantees, replay resistance

Equivalence / Privacy Properties: anonymity, unlinkability, observational equivalence,

What all can you NOT verify?

Cryptographic Strength & Computation

Implementation & Deployments

Real-time guarantees

Assumptions

What all can you NOT verify?

Cryptographic Strength & Computation

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Polynomial-time/probablistic adversaries

Quantitative Properties

User Intent or Semantic Meaning

Correctness of modelling or specifications.